

Civil Engineering

# Vectors, Matrices and Data Analysis

## Revision Questions

Use these questions for CAT and summative assessment preparation.

## Formula Practice

1. Apply the formula below in a technical example.

$$\vec{R} = \vec{A} + \vec{B}$$

2. Apply the formula below in a technical example.

$$AX = B$$

3. Apply the formula below in a technical example.

$$\det(A) = ad - bc$$

4. Apply the formula below in a technical example.

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

## Applied Practice

5. Resolve forces into components.
6. Solve simultaneous equations using matrices.
7. Calculate standard deviation of concrete test results.